

DATA MINING PROJECT

Master in Data Science and Advanced Analytics

NOVA Information Management School

Universidade Nova de Lisboa

Amazing International Airlines Inc. (AIAI) Business-Oriented Storytelling: From Analytics to Strategy

Group 25

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Fall/Spring Semester 2024-2025

CUSTOMER VALUE CONCENTRATION

The analysis indicated a high value concentration, with a small number of customers accounting for the majority of AIAI's profits. This shows the presence of a high Customer Lifetime Value (CLV) section, where CLV is the overall computed monetary worth of a customer relationship, as opposed to short-term income. Because AIAI's revenue is primarily reliant on this specific demographic, maintaining them is a strategic requirement. The company should prioritize protecting and growing this sector through focused retention campaigns, special awards, and premium-tier experiences that foster loyalty. However, losing even a small number of these valuable customers might have a big impact on overall profitability. To avoid this risk, AIAI might use predictive retention models to identify valued members who exhibit early indications of disengagement and intervene proactively to protect long-term value.

ZERO-INCOME CUSTOMERS

A substantial percentage of consumers indicated little to no income, presumably reflecting students or early-career travelers. Although their current income contribution is low, they demonstrate excellent engagement and have a high prospective lifetime value. AIAI may leverage on this opportunity by providing low-cost loyalty rewards and student-focused vacation packages to foster early brand engagement, allowing these clients to become long-term loyal members as their financial capacity grows.

ENGAGEMENT RECOVERY

Customer activity has increased dramatically after the epidemic began, exceeding pre-2020 levels. From a commercial standpoint, this is an ideal chance to reinforce customer habits and reactivate dormant members. AIAI might embrace this opportunity by temporarily lowering tier upgrade thresholds - for example, require fewer miles or points - or by delivering additional points and special benefits to members on the verge of achieving the next membership level.

GEOGRAPHIC CONCENTRATION

Over 75% of AIAI's customers are concentrated in Ontario, British Columbia, and Quebec, where engagement rates remain uniformly high (83–89%), indicating that current marketing investments are highly efficient but growth beyond these provinces is still underdeveloped. Therefore, AIAI should maintain strong investment in its core markets while pursuing low-cost digital acquisition strategies in smaller provinces to expand national presence efficiently.

BEHAVIORAL SEGMENTS

Flight pattern analysis identified two distinctive consumer groups: solo traveler and group or family travelers. This contrast highlights the differing priorities and needs between these groups: solo travelers frequently look for flexibility and personalized experiences, while group and family travelers typically value simplicity, coordinated planning, and shared benefits. To effectively meet these objectives, AIAI should create customized loyalty programs that provide flexible point redemption alternatives and personalized benefits for solitary travelers while giving bundled deals and family-oriented bonuses for group travelers. Furthermore, developing a specialized Group Booking Tool and launching family-focused loyalty programs will allow AIAI to capitalize on this high-margin although currently unexplored market niche, thereby enhancing both customer happiness and total income.

EXECUTIVE SUMMARY OF BUSINESS UNDERSTANDING

The analysis showed that AIAI's loyalty and profitability are strongly reliant on a small but extremely important client category. However, long-term development will be achieved by fostering emerging customer categories that have substantial future potential. Customer involvement has greatly improved, providing an excellent chance to reinforce new travel patterns and strengthen brand loyalty. Strategically, AIAI should find a balance between keeping and rewarding its top-tier clients and cultivating long-term relationships with emerging segments, while focusing the marketing budget to regions and channels that provide the most efficiency and effect.